David Talson

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Open to remote roles | Open to relocation

Role: Software & Systems Engineer | AI, Full-Stack, Computer Vision, and Product Integration

Professional Summary

Software and systems engineer with 6+ years of experience building and validating camera-to-cloud products across machine vision, embedded systems, AI, and full-stack SaaS.
Skilled in C++, Python, TypeScript, React, LLM APIs, computer vision, REST APIs, Docker, test automation, hardware bring-up, and CAN, Ethernet, TCP/UDP, UART, and serial communication.

Technical Skills

Languages: C++, Python, TypeScript, JavaScript, SQL, Shell Scripting
Full-Stack & Cloud: React, Vue.js, PostgreSQL, Supabase, REST APIs, Docker, Git, CI/CD
AI & Computer Vision: PyTorch, OpenCV, Object Detection, Image Classification, LLM APIs (OpenAI, Anthropic), RAG, LangChain, Model Context Protocol (MCP), Agentic Workflows
Machine Vision & Embedded: Qt6, GigE Vision, USB3 Vision, GenICam, NVIDIA Jetson, Embedded Linux, ROS2
Protocols & Device Integration: CAN Bus, Ethernet, TCP/UDP, UART, Serial Communication, eBUS SDK
Test, Hardware & Qualification: GoogleTest (GTest), Test Automation, Hardware Bring-Up, FPGA Validation, VPU Verification, EEPROM Programming, Wireshark, Oscilloscopes, EMI/EMC Testing, Environmental Qualification, MIL-STD-461G, MIL-STD-810H

Professional Experience

Founder & Product Engineer — Jun 2026 – Present
Eventxa (eventxa.com) — *Ottawa, ON*
Tech: TypeScript, React, Supabase, PostgreSQL, REST APIs, Stripe, LLM APIs (OpenAI, Anthropic), LangChain, MCP

- Built Eventxa, a serverless event-media SaaS for collecting, moderating, and displaying guest media at live events. The platform supports up to 300 concurrent guests and 3,000 uploads per event while keeping infrastructure costs below 5% of revenue through direct-to-storage uploads and client-side WebP compression.
- Created Smart Albums to group event photos in the organizer's browser using on-device face detection and 1024-dimensional embeddings. Added opt-in consent, immediate deletion on opt-out, and event-level data purging so biometric data is never sent to a server.
- Implemented Eventxa's four-tier subscription flow, from Stripe Checkout and webhook fulfillment to account access and plan entitlements. Built the public REST API, custom QR-code generator, live photo wall, retention controls, and self-service account and data-deletion flows.

Software Developer — Apr 2022 – Mar 2026
Pleora Technologies — *Ottawa, ON*
Tech: C++, Qt 6, eBUS SDK, GigE Vision, USB3 Vision, GenICam, Serial Communication

- Built capture and recording workflows for eBUS Player 7 in Qt 6. Unlike eBUS Player 6, users can capture or record from up to 4 connected GigE Vision or USB3 Vision devices at the same time, with selectable BMP, TIFF, PNG, raw-binary, and video formats (Docs).
- Added the Event Monitor and console log features to help users investigate device and stream issues. Users can track GenICam updates, streaming statistics, and dropped or missed buffers for one device or all connected devices, then save the logs for troubleshooting (Docs).

- Worked with the eBUS SDK and application team to automate buffer memory-address assignment. The validation checks each requested address range before allocation, preventing the silent overlap failures possible with manual address and length entry. Built a configurable serial-communication bridge for device setup and validation.

Graduate Researcher — Project Report May 2023 – Aug 2023
Smart Connected Vehicles Innovation Centre *Ottawa, ON*
Tech: Python, MATLAB/Simulink, QUARC, Faster R-CNN, OptiTrack, Computer Vision

- Designed and implemented an autonomous Quanser QDrone system that searched a grid, detected targets from live video with Faster R-CNN, estimated each target's real-world position, and landed on the detected target (Demo).
- Integrated and tested the real-time control pipeline in MATLAB/Simulink and QUARC with OptiTrack 6-DoF localization, host-to-drone video streaming, FIFO buffering, and regression-based coordinate estimation. Reached 0.95 mAP for target detection and up to 0.976 R^2 for position prediction.

Intermediate Product Verification Engineer Jan 2022 – Mar 2022
Pleora Technologies *Ottawa, ON*
Tech: VPU Verification, Test Automation, EMI/EMC, MIL-STD-461G, MIL-STD-810H

- Served as the primary technical owner for RuggedCONNECT video processing units from prototype validation through production release. Wrote assembly, build, and test instructions, then supported manufacturing during the R&D-to-production transition.
- Led VPU performance verification during military qualification testing at MBP, TÜV SÜD, and the Canadian Space Agency. Automated recurring procedures and supported EMI/EMC, shock and vibration, temperature, and humidity testing to improve test consistency and turnaround.

Junior Product Verification Engineer Apr 2021 – Dec 2021
Pleora Technologies *Ottawa, ON*
Tech: C++, GoogleTest, NVIDIA Jetson TX2i, Embedded Linux, CAN Bus, Ethernet

- Led end-to-end bring-up, verification, and automated testing of Jetson TX2i-based video processing units across Linux BSP deployment, EEPROM provisioning, CAN/Ethernet validation, and AI image workflows, improving test accuracy and reducing test time by 30%.

Product Verification & Test Automation Mar 2020 – Apr 2021
Pleora Technologies *Ottawa, ON*
Tech: Docker, CI/CD, Shell Scripting, Linux, Jenkins, FPGA Validation

- Built Docker-based automated test environments and shell-scripted SDK installation and upgrade workflows for CI/CD validation, while also programming and testing FPGA-based production devices under ESD-controlled verification processes.

Earlier Experience
Hardware Co-op, Test Engineering, Sanmina Corporation May 2019 – Dec 2019
Embedded Software Designer Co-op, Pleora Technologies Jan 2019 – Apr 2019
Technical Support Specialist, Carleton University Sep 2018 – Dec 2018
Electrical Engineer Co-op, Measurements International May 2018 – Aug 2018

Education

Master of Engineering, Electrical and Computer Engineering
University of Ottawa — Concentration in Applied Artificial Intelligence
Graduated: 2024 | GPA: 8.0/10.0 (A-)

Bachelor of Engineering, Electrical Engineering
Carleton University
Graduated: 2020